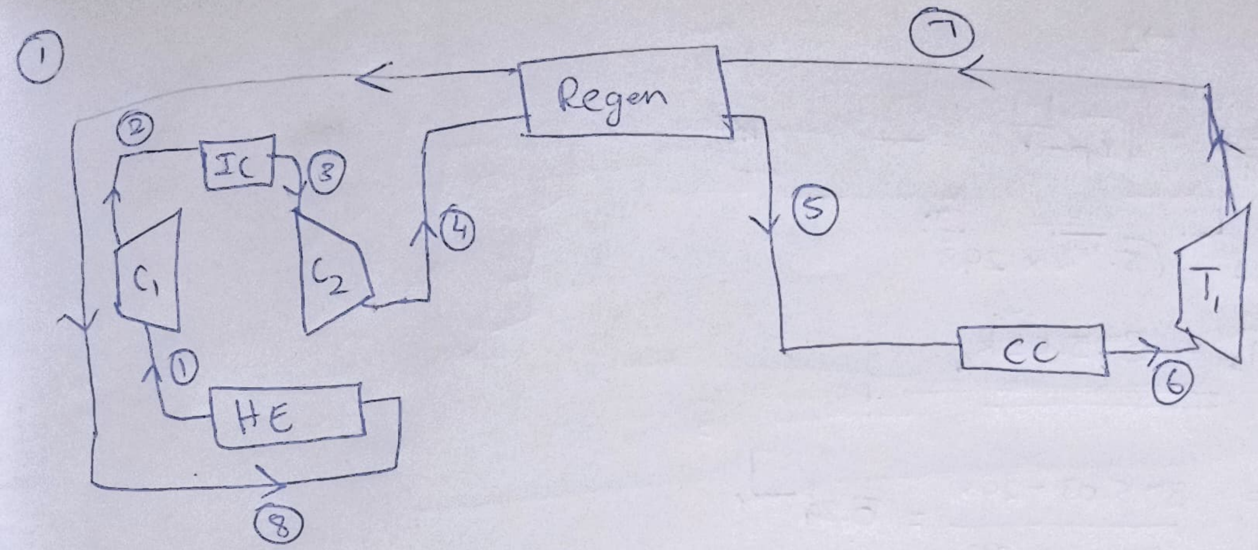
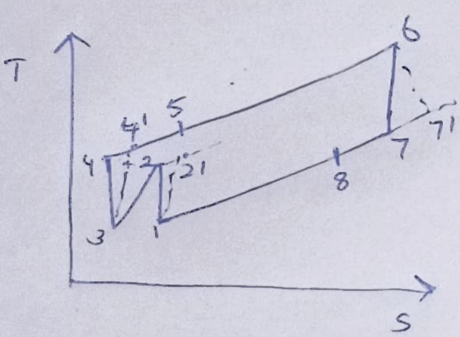




$r_p = 5$

$T_6 = 1100°C = 1373K$

$T_1 = T_3 = 25°C = 298K$

$\eta_T = 0.86$

$\eta_c = 0.89$

$\varepsilon = 0.70$

Overall pressure ratio:

$$r_p = \frac{P_4}{P_1} = \frac{P_4}{P_3} = 5$$

$$\frac{P_2}{P_1} = \frac{P_4}{P_3} = \sqrt{5} = 2.24 = x$$

$$\eta_{c_1} = \frac{h_2 - h_1}{h_2' - h_1} = \frac{(T_2 - T_1)}{T_2' - T_1} = 0.89$$

$$\eta_{c_2} = \frac{h_4 - h_3}{h_4' - h_3} = \frac{T_4 - T_3}{T_4' - T_3} = 0.89$$

$$\eta_{T_1} = \frac{T_6 - T_7'}{T_6 - T_7} = 0.86$$

Process $1 \to 2$:

$$\frac{T_2}{T_1} = \sqrt{r_p}^{\frac{\gamma-1}{\gamma}}$$

$$T_2 = \sqrt{5}^{\frac{0.4}{1.4}} \times 298$$

$$T_2 = 375.03K$$

$$\eta_{c_1} = \frac{375.03-298}{\underset{T_2'}{\cancel{375.03}}-\cancel{298}} = 0.89$$

$$T_2' = 384.55\,K$$

Process $3 \to 4$:

$$\frac{T_4}{T_3} = \sqrt{r_p}^{\frac{\gamma-1}{\gamma}}$$

$$T_4 = \sqrt{5}^{\frac{0.4}{1.4}} \times 298$$

$$T_4 = 375.03K = T_2$$

~~He~~ Since, $\eta_{c_1} = \eta_{c_2}$, $T_4 = T_2 \to \sqrt{r_p} = \text{same}$

$$[T_2' = T_4' = 384.55K]$$

$$W_{c_1} = mC_p(T_2' - T_1)$$

$$= 1.005 \times (384.55 - 298)$$

$$W_{c_1} = 86.98\,kJ/kg = W_{c2}$$

Process $6 \to 7$:

$$\frac{T_6}{T_7} = r_p^{\frac{0.4}{1.4}}$$

$$T_7 = \frac{T_6}{r_p^{0.4/1.4}} = \frac{1373}{5^{\frac{0.4}{1.4}}}$$

$$T_7 = 866.89\,K$$

$$\eta_T = \frac{T_6 - T_7'}{T_6 - T_7} = 0.86$$

$$T_7' = -0.86(1373 - 866.89) + 1373$$

$$T_7' = 937.75\,K$$

$$w_T = C_p (T_6 - T_7')$$
$$= 1.005(1373 - 937.75)$$
$$w_T = 437.43\,\cancel{KW}\,J/Kg$$

$$W_{Net} = W_T - 2W_c$$
$$= 437.43 - 2(86.98)$$
$$W_{net} = 263.47\,kJ/kg$$

$$Q_{in} = \cancel{M}\,C_p (T_6 - T_5)$$
$$= 1.005(1373 - T_5)$$

$$\varepsilon_p = \frac{T_5 - T_4'}{T_7' - T_4'} = 0.70$$

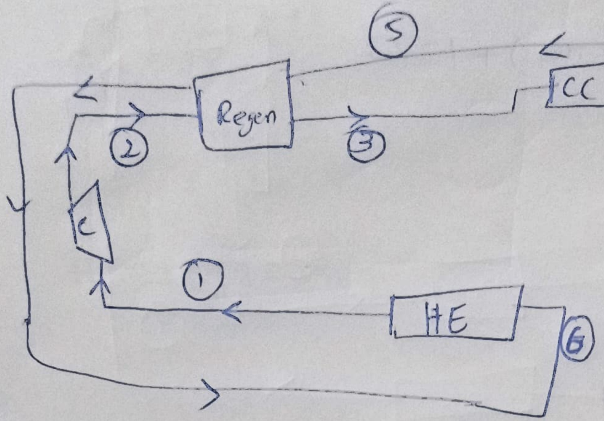
$$\frac{T_5 - \cancel{374}\,375.03}{937.75 - 384.55} = 0.70$$

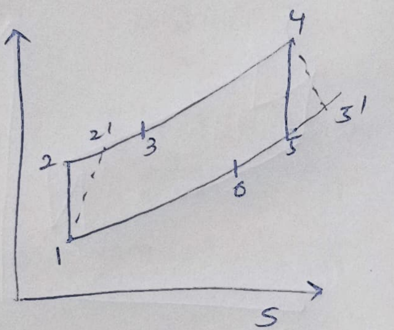
$$T_5 = 762.27\,K$$

$$Q_{in} = 1.005(1373 - 762.27)$$
$$Q_{in} = 613.78\,KJ/kg$$

$$\eta = \frac{\cancel{Q_{in}}\,W_{net}}{Q_{in}} = \frac{263.47}{613.78} = 0.429 = 42.9\,\%$$

(2)





$P_1 = P_5 = 1 bar$

$P_2 = P_4 = 5.2 bar$

$T_1 = 25°C = 298K$

$T_4 = 1000°C = 1273K$

$\eta_T = 0.88$, $\eta_c = 0.84$, $\varepsilon_p = 0.66$

Process 1→2:

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_2 = \left(\frac{5.2}{1}\right)^{\frac{0.4}{1.4}} \times 298$$

$$\underline{T_2 = 477.297 K}$$

$$\eta_c = \frac{T_2 - T_1}{T_2' - T_1} = 0.84$$

$$T_2' = \frac{477.297 - 298}{0.84} + 298$$

$$T_2' = \underline{511.45 K}$$

Process 4 → 5

$$\frac{T_4}{T_5} = \left(\frac{P_4}{P_5}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_5 = \frac{1273}{(5.2)^{0.4/1.4}}$$

$$T_5 = 794.797\ K$$

$$\eta_T = \frac{T_4 - T_5'}{T_4 - T_5} = \cancel{0.66}\ 0.88$$

$$T_5' = -0.88(1273 - 794.797) + 1273$$

$$T_5' = 852.18\ K$$

$$\varepsilon_p = \frac{T_3 - T_2'}{T_5' - T_2'} = 0.66$$

$$T_3 = 0.66(852.18 - \overset{511.45}{\cancel{477.297}}) + 511.45$$

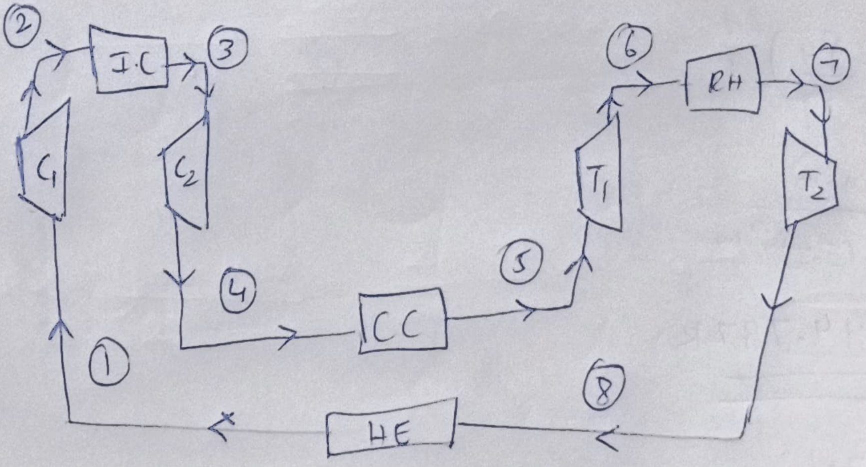
$$T_3 = 736.33\ K$$

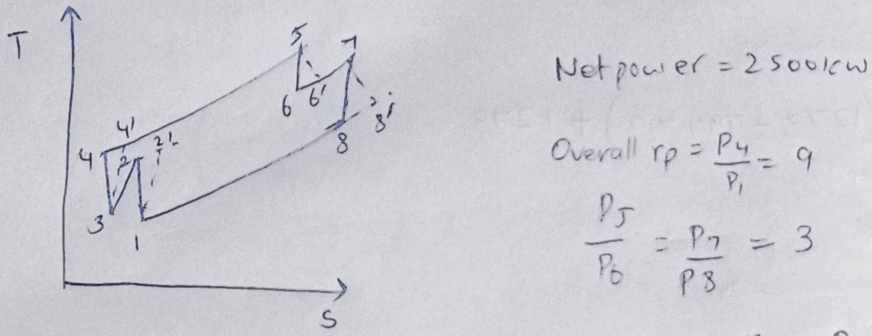
$$\eta = \frac{W_{net}}{Q_{in}} = \frac{W_T - W_c}{Q_{in}} = \frac{\cancel{C_p}(T_4 - T_5') - \cancel{C_p}(T_2' - T_1)}{\cancel{C_p}(T_4 - T_3)}$$

$$\eta = \frac{T_4 - T_5' - T_2' + T_1}{T_4 - T_3} = \frac{1273 - 852.18 - 511.45 + 298}{1273 - 736.33}$$

$$\eta = 0.386 = 38.6\%$$

(3)





Net power = 2500kW

Overall $r_p = \frac{P_4}{P_1} = 9$

$\frac{P_5}{P_6} = \frac{P_7}{P_8} = 3$

$\eta_c = 0.78, \eta_t = 0.82$

$T_5 = 970°C = T_7 = 1243K$

$T_1 = 25°C = 298K$

$\frac{P_4}{P_1} = 9$

$\frac{P_2}{P_1} = \frac{P_4}{P_3} = \sqrt{9} = 3$

Process 1→2

$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$

$T_2 = (3)^{\frac{0.4}{1.4}} \times 298$

$T_2 = 407.88\ K$

Since $\frac{P_4}{P_3} = \frac{P_2}{P_1}$ and $\eta_{c_1} = \eta_{c_2}$

$\hookrightarrow T_2 = T_4$

$T_2' = T_4'$

$T_1 = T_3$

$$0.78 = \frac{T_2 - T_1}{T_2' - T_1}$$

$$T_2' = \frac{407.88 - 298}{0.78} + 298$$

$$T_2' = 438.87K$$

Process 5→6

$$\frac{T_5}{T_6} = \left(\frac{P_5}{P_6}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_6 = \frac{1243}{(3)^{0.4/1.4}}$$

$$T_6 = 908.14K$$

$$0.82 = \frac{T_5 - T_6'}{T_5 - T_6}$$

$$T_6' = T_5 - 0.82(T_5 - T_6)$$

$$= 1243 - 0.82(1243 - 908.14)$$

$$T_6' = 968.41K$$

$$W_c = 1.005(T_2' - T_1)$$

$$= 1.005(438.87 - 298)$$

$$W_c = 141.57 \, kJ/kg$$

$$W_T = 1.005(T_5 - T_6')$$

$$W_T = 1.005(1243 - 968.41)$$

$$W_T = 275.96 \, kJ/kg$$

$$W_{net} = 2W_T - 2W_C$$

$$= 2(W_T - W_C)$$

$$W_{net} = 2(275.96 - 141.57)$$

$$W_{net} = 268.78 \, kJ/kg$$

$$\bar{m} = \frac{Power}{W_{net}} = \frac{2500}{\cancel{290.62}\ 268.78} = 9.30\ kg/s$$

$$W_R = \frac{2(W_T - W_C)}{2W_T} = \frac{268.78}{2 \times (275.96)} = 0.487 = 48.7\%$$

↳ work ratio

$$Q_{in} = 1.005(T_5 - T_4')$$

$$= 1.005(1243 - 438.87)$$

$$Q_{in} = 808.15\ kJ/kg$$

$$\eta = \frac{W_{net}}{Q_{in}} = \frac{268.78}{808.15} = 0.333 = 33.3\%$$